

This report analyzes the performance of five parent trading orders (V001–V005), breaking down execution tactics, market impact, and benchmark performance against real-time market conditions.

Step 1: Data Gathering & Preprocessing

First, I consolidated raw data from four core sources:

- **Parent Orders:** The full-sized target orders to execute
- **Child Orders:** The split, market-sent order slices
- **Trades:** All executed market transactions
- **Quotes:** Real-time bid/ask price updates throughout the trading day

I standardized timestamps by merging separate date/time columns into a unified timestamp for every record, ensuring precise alignment of our orders with split-second market conditions.

Step 2: End-of-Day Time Standardization

To ensure fair MOC (Market on Close) performance measurement, I applied a time-grouping rule: all trades recorded after 14:57 were consolidated and timestamped to 15:00, aligning with market close benchmarks.

Step 3: Core Metric Calculation

For each parent order, I calculated key execution metrics during its active trading window:

- **Notional Value:** Total order size (in millions of CNY)
- **ADV% (Trading Speed):** Our order volume as a percentage of total market volume during the window (higher = larger market footprint)
- **Spread (bps):** Average bid-ask spread during execution (wider = higher execution friction)

Step 4: Benchmark Performance Analysis

I measured execution cost (in basis points, bps) against industry-standard benchmarks to evaluate performance:

- **Open Price:** Day's opening price
 - **Arrival Price:** Market price immediately before order initiation
 - **iVWAP (Interval VWAP):** Volume-weighted average price of all market trades during our execution window
 - **Close Price:** Day's closing price
 - **PWPS (Participation Weighted Price):** Simulated benchmark for 5% market volume participation throughout the window
 - **MOO% / MOC%:** Percentage of order executed at Market on Open / Market on Close
 - **Passive% / Aggressive%:** Percentage of trades executed at bid (passive, waiting for market) vs. ask (aggressive, crossing spread)
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Order-by-Order Performance Breakdown

Order V001

- **Basics:** 15.89M CNY notional, 0.90% ADV, 8.60 bps average spread
- **Performance:** Underperformed open/arrival price (-30.88 bps vs. both), but vastly outperformed close price (+121.47 bps), indicating the market moved against us later in the day, making early execution favorable. Beat PWPS benchmark (+0.76 bps).
- **Tactics:** Highly aggressive execution (64.53% aggressive, 32.29% passive), with 3.19% of volume executed at MOO, 0% at MOC.

Order V002

- **Basics:** 23.39M CNY notional, 0.79% ADV, 3.98 bps average spread
- **Performance:** Strong outperformance vs. arrival price (+25.83 bps), slight underperformance vs. iVWAP (-4.19 bps) and close price (-13.14 bps).
- **Tactics:** Aggressive execution (66.69% aggressive, 32.85% passive), 0% MOO/MOC execution.

Order V003

- **Basics:** 4.47M CNY notional (smallest order), 0.35% ADV, 7.32 bps average spread
- **Performance:** Significant underperformance vs. open/arrival price (-69.38 bps vs. both) and close price (-112.44 bps), with -21.57 bps vs. PWPS, driven by adverse market price movement during execution.
- **Tactics:** Balanced strategy (48.27% passive, 41.79% aggressive), with 9.94% of volume executed at MOO, 0% at MOC.

Order V004

- **Basics:** 28.49M CNY notional, 1.82% ADV, 4.06 bps average spread
- **Performance:** Strong execution, outperforming open/arrival price (+3.77 bps vs. both) and PWPS (+4.96 bps), with minor underperformance vs. iVWAP (-1.66 bps) and close price (-10.11 bps).
- **Tactics:** Slightly aggressive, balanced execution (52.20% aggressive, 43.32% passive), with 3.27% of volume executed at MOC, 0% at MOO.

Order V005

- **Basics:** 33.14M CNY notional (largest order), 2.27% ADV, 7.16 bps average spread
- **Performance:** Extreme underperformance vs. open/arrival price (-149.28 bps vs. both), but strong outperformance vs. iVWAP (+17.86 bps) and PWPS (+93.13 bps), reflecting high market volatility/trending conditions during execution.
- **Tactics:** Highly passive strategy (62.38% passive, 36.87% aggressive), with 0.73% MOO and 3.00% MOC execution.

Aggregate (All Orders)

- **Basics:** 105.39M CNY total notional, 1.53% average ADV, 5.84 bps average spread
- **Performance:** -51.71 bps vs. open, -47.79 bps vs. arrival, +3.58 bps vs. iVWAP, -33.71 bps vs. close, +26.91 bps vs. PWPS
- **Tactics:** Balanced overall execution (45.53% passive, 52.02% aggressive), with 1.13% MOO and 1.83% MOC execution.

Key Insights & Strategic Takeaways

1. Order Size Drives Execution Strategy

Small-to-medium orders (V001, V002) used aggressive execution (~65% aggressive) to secure fast fills, while the largest order (V005) shifted to a highly passive strategy (~62% passive) to minimize market impact and avoid moving prices against the position. This algorithmic logic aligns with best practices for large-block trading.

2. iVWAP as the Most Reliable Benchmark

Open/close price benchmarks are distorted by full-day market movement, while iVWAP (interval VWAP) directly measures execution against concurrent market activity. Across all orders, iVWAP performance ranged from -17 bps to +18 bps, confirming our algorithms consistently track real-time market conditions efficiently.

3. Market Impact for Large, Fast Trades

High-ADV orders (V004, V005, 1.8–2.27% of market volume) showed extreme divergence in arrival price performance, even with passive execution for V005. This highlights the challenge of executing large positions quickly: order size itself creates market impact, making arrival price a poor standalone metric for large-block trades.

Conclusion

This analysis validates that our automated trading strategies perform logically and efficiently in live market conditions. Algorithms adjust aggression based on order size, prioritize passive execution for large positions to reduce market impact, and consistently deliver fair execution relative to concurrent market prices (iVWAP). While start-of-day benchmark performance varies with market volatility, our core execution quality remains strong across all order sizes.

Gemini Pro 3 was used to formalize the report and refine grammar.